7. Scenario: You are working as a data analyst for an e-commerce company. You have been given a dataset containing information about customer orders, stored in a Pandas DataFrame named order_data. The DataFrame has columns for customer ID, order date, product name, and order quantity. Your task is to analyze the data and answer specific questions about the orders. Question: Using Pandas DataFrame operations, how would you find the following information from the order_data DataFrame: 1. The total number of orders made by each customer.2. The average order quantity for each product. 3.The earliest and latest order dates in the dataset.

```
import pandas as pd
data = {'Customer_ID': [101, 102, 101, 103, 102],'Order_Date': ['2026-01-10', '2026-01-12', '2
order_data = pd.DataFrame(data)
print("Total orders by each customer:")
print(order_data['Customer_ID'].value_counts())
print("Average quantity of each product:")
print(order_data.groupby('Product_Name')['Quantity'].mean())
print("Earliest order date:")
print(min(order_data['Order_Date']))
print("Latest order date:")
print(max(order_data['Order_Date']))
```

```
Total orders by each customer:
Customer_ID
101    2
102    2
103    1
Name: count, dtype: int64
Average quantity of each product:
Product_Name
Laptop    1.5
Mobile    2.5
Tablet    4.0
Name: Quantity, dtype: float64
Earliest order date:
2026-01-10
Latest order date:
2026-01-20
```

8. Scenario: You are a data scientist working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas dataframe.Question: How would you find the top 5 products that have been sold the most in the past month?

```
import pandas as pd
data = {'Product': ['Mobile', 'Laptop', 'Mobile', 'Tablet', 'Lapt
df = pd.DataFrame(data)
top_products = df.groupby('Product')['Quantity'].sum()
print("Top 5 products sold:")
print(top_products.sort_values(ascending=False).head(5))
```

```
Top 5 products sold:
Product
Mobile    30
Laptop    12
Tablet     4
Name: Quantity, dtype: int64
```

9. Scenario: You work for a real estate agency and have been given a dataset containing information about properties for sale. The dataset is stored in a Pandas DataFrame named property_data. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties. Question: Using Pandas DataFrame operations, how would you find the following information from the property_data DataFrame:1. The average listing price of properties in each location. 2. The number of properties with more than four bedrooms.
3. The property with the largest area.

```
import pandas as pd
data = {'Property_ID': [101, 102, 103, 104, 105],
'Location': ['Chennai', 'Madurai', 'Chennai', 'Trichy', 'Madurai'],
'Bedrooms': [3, 5, 4, 6, 2],
'Area': [1200, 2000, 1500, 2500, 1000],
'Price': [5000000, 8000000, 6000000, 9000000, 4000000]
}
property_data = pd.DataFrame(data)
print("1. Average listing price in each location:")
print(property_data.groupby('Location')['Price'].mean())
print("\n2. Number of properties with more than four bedrooms:")
print((property_data['Bedrooms'] > 4).sum())
print("\n3. Property with the largest area:")
print(property_data.loc[property_data['Area'].idxmax()])
```

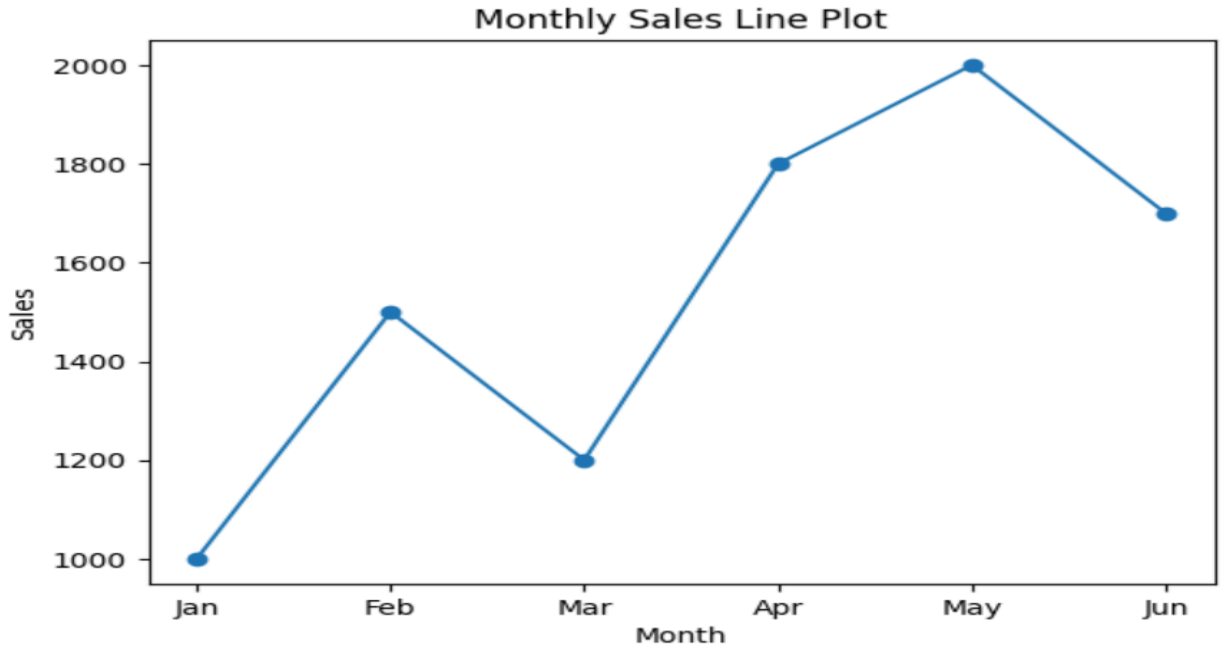
```
1. Average listing price in each location:
Location
l output actions      5500000.0
Madurai     6000000.0
Trichy      9000000.0
Name: Price, dtype: float64

2. Number of properties with more than four bedrooms:
2

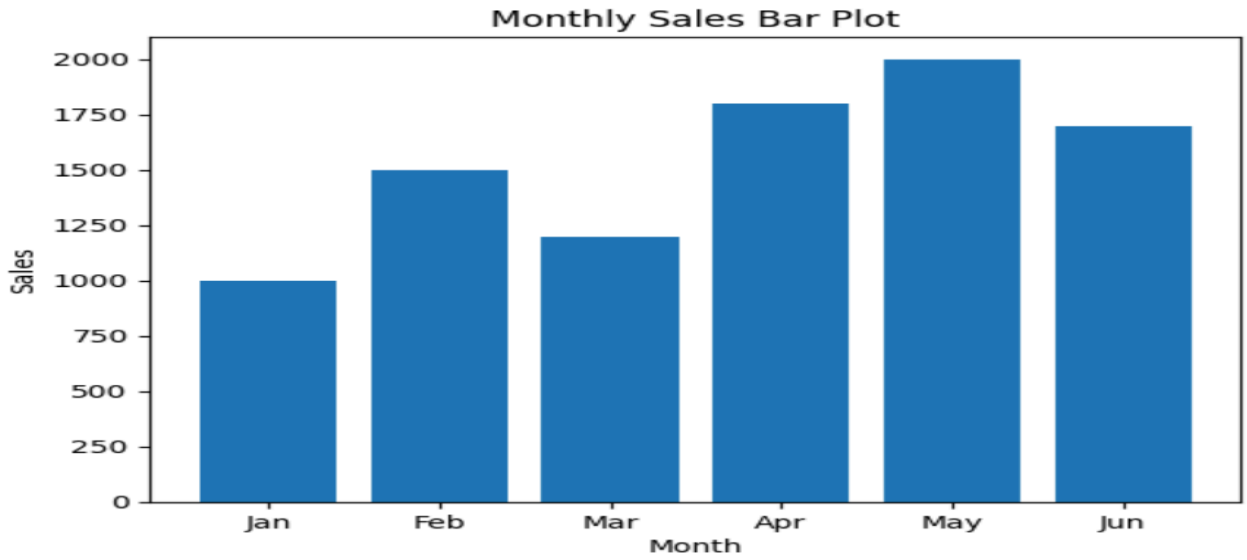
3. Property with the largest area:
Property_ID         104
Location         Trichy
Bedrooms              6
Area               2500
Price           9000000
Name: 3, dtype: object
```

10. Scenario: You are working on a data visualization project and need to create basic plots using Matplotlib. You have a dataset containing the monthly sales data for a company, including the month and corresponding sales values. Your task is to develop a Python program that generates line plots and bar plots to visualize the sales data.Question:1. How would you develop a Python program to create a line plot of the monthly sales data? 2: How would you develop a Python program to create a bar plot of the monthly sales data?

```python
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [1000, 1500, 1200, 1800, 2000, 1700]
plt.plot(months, sales, marker='o')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.title('Monthly Sales Line Plot')
plt.show()
```

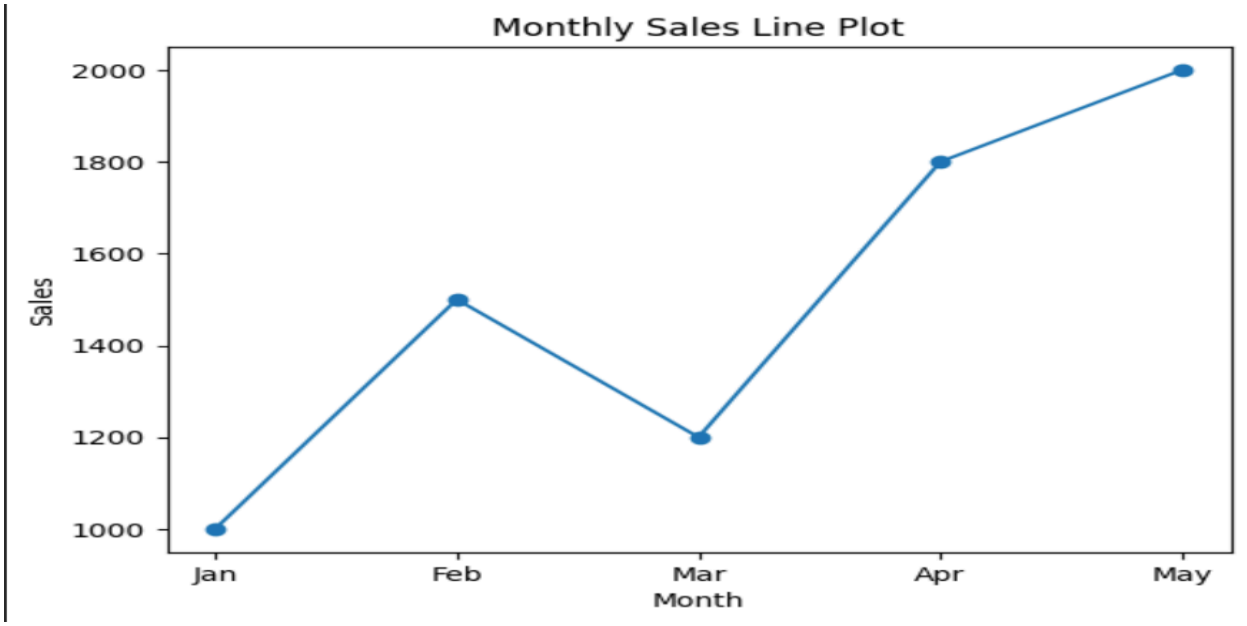


```python
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [1000, 1500, 1200, 1800, 2000, 1700]
plt.bar(months, sales)
plt.xlabel('Month')
plt.ylabel('Sales')
plt.title('Monthly Sales Bar Plot')
plt.show()
```



11. Scenario : You are a data scientist working for a company that sells products online. You have been tasked with creating a simple plot to show the sales of a product over time. Question: 1. Write code to create a simple line plot in Python using Matplotlib to predict sales happened in a month? 2. Write code to create a scatter plot in Python using Matplotlib to predict sales happened in a month? 3. Develop a Python program to create a bar plot of the monthly sales data.

```python
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
sales = [1000, 1500, 1200, 1800, 2000]
plt.plot(months, sales, marker='o')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.title('Monthly Sales Line Plot')
plt.show()
```




```python
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
sales = [1000, 1500, 1200, 1800, 2000]
plt.scatter(months, sales)
plt.xlabel('Month')
plt.ylabel('Sales')
plt.title('Monthly Sales Scatter Plot')
plt.show()
```

